



7772 - Interstellar PAHs: separating the neutrals from the ions

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
NGC 2023				
	2	NGC2023-MIRI-MRS	MIRI Medium Resolution Spectroscopy	(1) NGC2023
	17	NGC2023-BKG-MIRI-MRS	MIRI Medium Resolution Spectroscopy	(9) NGC2023-BKG
NGC 1333				
	4	NGC1333-MIRI-MRS	MIRI Medium Resolution Spectroscopy	(2) NGC1333
	18	NGC1333-BKG-MIRI-MRS	MIRI Medium Resolution Spectroscopy	(10) NGC1333-BKG
vdb133				
	5	vdb133-NIRspec-IFU	NIRSpec IFU Spectroscopy	(3) VDB133
	6	vdb133-BKG-NIRspec-IFU	NIRSpec IFU Spectroscopy	(4) VDB133-BKG
	7	vdb133-MIRI-MRS	MIRI Medium Resolution Spectroscopy	(3) VDB133

Folder	Observation	Label	Observing Template	Science Target
	8	vdb133-BKG-MIRI-MRS	MIRI Medium Resolution Spectroscopy	(4) VDB133-BKG
vdb111				
	9	vdb111-NIRspec-IFU	NIRSpec IFU Spectroscopy	(5) VDB111
	10	vdb111-BKG-NIRspec-IFU	NIRSpec IFU Spectroscopy	(6) VDB111-BKG
	11	vdb111-MIRI-MRS	MIRI Medium Resolution Spectroscopy	(5) VDB111
	12	vdb111-BKG-MIRI-MRS	MIRI Medium Resolution Spectroscopy	(6) VDB111-BKG
vdb101				
	13	vdb101-NIRspec-IFU	NIRSpec IFU Spectroscopy	(7) VDB101
	14	vdb101-BKG-NIRspec-IFU	NIRSpec IFU Spectroscopy	(8) VDB101-BKG
	15	vdb101-MIRI-MRS	MIRI Medium Resolution Spectroscopy	(11) VDB101-visit-15-copy
	16	vdb101-BKG2-MIRI-MRS	MIRI Medium Resolution Spectroscopy	(12) VDB101-BKG-visit-15-copy

ABSTRACT

JWST observations are dominated by strong emission bands at 3.3, 6.2, 7.7, 8.6, and 11.2 μm , referred to as the Aromatic Infrared Bands (AIBs) and generally attributed to polycyclic aromatic hydrocarbons (PAHs). PAHs play a key role in several physical and chemical processes highly relevant for large-scale astrophysical processes such as star and planet formation and galaxy evolution.

These physical and chemical processes strongly depend on the PAH charge distribution. The PAH charge distribution also governs the spectral characteristics of the AIB emission spectra. However, we are currently unable to derive a consistent quantitative value for the PAH ionization fraction from observations. This in turn affects the quantitative role PAHs play in the Universe, e.g. an incorrect PAH ionization fraction strongly influences the heating rate of the neutral ISM.

We request NIRSpec IFU and MIRI MRS observations of a sample of reflection nebulae illuminated by central stars of a range of spectral types. This will include the first good quality, medium spectral resolution spectrum of PAH emission from a single charge state, i.e. neutral PAHs. We will separate the spectroscopic properties of neutral and cationic PAHs and systematically calibrate these in terms of the radiation field. This program will thus provide observational templates of purely neutral and purely cationic AIB emission as well as the molecular properties of their carriers. We will

directly quantify the PAH ionization fraction from the observations which will allow us to assess methods currently employed to derive the PAH ionization fraction and to quantitatively validate PAH models and thus the PAH hypothesis!

OBSERVING DESCRIPTION

To achieve the science goals of this project we will use the NIRSpec and MIRI MRS IFUs to study a sample of 5 reflection nebulae: vdB 101, vdB 111, vdB 133, NGC 2023, and NGC 1333. For NGC2023 and NGC1333, we request MIRI MRS observations at the location of existing NIRSpec IFU observations.

We will use NIRSpec IFU with the G395M/F290LP disperser-filter combination to cover the 2.87-5.27 wavelength range at a resolution of ~ 1000 , and all four MIRI MRS IFU channels and their respective sub-bands.

We employ the “4-Point” Dither pattern for NIRSpec IFU and the “Extended Source, 4-Point ALL” dither pattern for the MIRI MRS observations.

For MIRI MRS, we will use the FASTR1 readout pattern for the brightest sources (NGC 1333, NGC 2023) and the SLOWR1 readout pattern for the fainter sources (vdB 133, vdB 101 and vdB 111). For NIRSpec IFU, we will use the NRSIRS2 readout pattern for vdB 133, vdB 101 and vdB 111.

We request dedicated off-source background exposures in both instruments using the same configuration as the on-source observation. We will perform Leakcal exposures for all the NIRSpec IFU observations, except for the background dedicated observations. The Leakcal exposures will be set to the exact exposure parameters and configuration as the main science regions.

Sources vdb 101 and vdb 111 have been assessed as "Background Limited observations" and therefore a ``Background-Limited" special requirement is selected with the "Background no more than the 10th percentile above minimum" option for both NIRSpec and MIRI MRS IFU observations, based on the guidelines in the JWST documentation.

Following JWST documentation recommendations, we perform MIRI-MRS simultaneous imaging to improve the astrometric solution of individual MRS exposures.

For sources where schedulability based on the Visit Planner allowed, we set a V3 PA requirement between the NIRSpec and MIRI-MRS observations, to ensure maximum overlap between their field of views.

Proposal 7772 - Targets - Interstellar PAHs: separating the neutrals from the ions

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	NGC2023	RA: 05 41 37.8600 (85.4077500d) Dec: -02 16 50.37 (-2.28066d) Equinox: J2000 <i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>Category=ISM</i> <i>Description=[Reflection nebulae]</i> <i>Extended=YES</i>	Epoch of Position: 2015.5	
	(2)	NGC1333	RA: 03 29 10.4000 (52.2933333d) Dec: +31 21 51.12 (31.36420d) Equinox: J2000 <i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>Category=ISM</i> <i>Description=[Reflection nebulae]</i> <i>Extended=YES</i>	Epoch of Position: 2015.5	
	(3)	VDB133	RA: 20 31 4.2115 (307.7675479d) Dec: +36 57 22.67 (36.95630d) Equinox: J2000 <i>Comments:</i> <i>Category=ISM</i> <i>Description=[Nebulae, Reflection nebulae]</i> <i>Extended=YES</i>		
	(4)	VDB133-BKG	RA: 20 31 16.7318 (307.8197158d) Dec: +36 50 42.01 (36.84500d) Equinox: J2000 <i>Comments:</i> <i>Category=Calibration</i> <i>Description=[Telescope/sky background]</i> <i>Extended=NO</i>		
	(5)	VDB111	RA: 17 18 52.3455 (259.7181063d) Dec: +06 04 36.98 (6.07694d) Equinox: J2000 <i>Comments:</i> <i>Category=ISM</i> <i>Description=[Nebulae, Reflection nebulae]</i> <i>Extended=YES</i>		
	(6)	VDB111-BKG	RA: 17 19 11.4937 (259.7978904d) Dec: +06 05 47.71 (6.09659d) Equinox: J2000 <i>Comments:</i> <i>Category=Calibration</i> <i>Description=[Telescope/sky background]</i> <i>Extended=NO</i>		
	(7)	VDB101	RA: 16 19 2.2506 (244.7593775d) Dec: -20 14 32.10 (-20.24225d) Equinox: J2000 <i>Comments:</i> <i>Category=ISM</i> <i>Description=[Nebulae, Reflection nebulae]</i> <i>Extended=YES</i>		

Proposal 7772 - Targets - Interstellar PAHs: separating the neutrals from the ions

(8)	VDB101-BKG	RA: 16 19 36.0675 (244.9002813d) Dec: -20 18 47.83 (-20.31329d) Equinox: J2000
Comments: Category=Calibration Description=[Telescope/sky background] Extended=NO		
(9)	NGC2023-BKG	RA: 05 42 6.8400 (85.5285000d) Dec: -02 14 34.30 (-2.24286d) Equinox: J2000
Comments: Category=Calibration Description=[Telescope/sky background] Extended=NO		
(10)	NGC1333-BKG	RA: 03 29 21.5100 (52.3396250d) Dec: +31 20 50.90 (31.34747d) Equinox: J2000
Comments: Category=Calibration Description=[Telescope/sky background] Extended=NO		
(11)	VDB101-visit-15-copy	RA: 16 19 2.2506 (244.7593775d) Dec: -20 14 32.10 (-20.24225d) Equinox: J2000
Comments: Category=ISM Description=[Nebulae, Reflection nebulae] Extended=YES		
(12)	VDB101-BKG-visit-15-copy	RA: 16 20 51.8165 (245.2159021d) Dec: -20 27 42.07 (-20.46169d) Equinox: J2000
Comments: Category=Calibration Description=[Telescope/sky background] Extended=NO		

Proposal 7772 - Observation 2 - Interstellar PAHs: separating the neutrals from the ions

Observation	Proposal 7772, Observation 2: NGC2023-MIRI-MRS												Mon Aug 04 19:00:20 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[NGC2023-BKG-MIRI-MRS (Obs 17)]												
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(1)	NGC2023	RA: 05 41 37.8600 (85.4077500d) Dec: -02 16 50.37 (-2.28066d) Equinox: J2000					Epoch of Position: 2015.5					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	Category=ISM												
	Description=[Reflection nebulae]												
Extended=YES													
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	F560W	All MRS				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction			
	1	4-Point					EXTENDED SOURCE			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1		IMAGER	F770W	FASTR1	5	3	1	Dither 1	4	12	188.703	
	1	SHORT(A)	MRSLONG		FASTR1	22	1	1	Dither 1	4	4	244.204	213681
	1	SHORT(A)	MRSSHORT		FASTR1	22	1	1	Dither 1	4	4	244.204	213681
	2		IMAGER	F1000W	FASTR1	5	3	1	Dither 1	4	12	188.703	
	2	MEDIUM(B)	MRSLONG		FASTR1	22	1	1	Dither 1	4	4	244.204	213681
	2	MEDIUM(B)	MRSSHORT		FASTR1	22	1	1	Dither 1	4	4	244.204	213681
	3		IMAGER	F1130W	FASTR1	5	3	1	Dither 1	4	12	188.703	
	3	LONG(C)	MRSLONG		FASTR1	22	1	1	Dither 1	4	4	244.204	213681
	3	LONG(C)	MRSSHORT		FASTR1	22	1	1	Dither 1	4	4	244.204	213681

Proposal 7772 - Observation 2 - Interstellar PAHs: separating the neutrals from the ions

Special Requirements	Sequence Observations 2, 17, Non-interruptible
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Proposal 7772 - Observation 17 - Interstellar PAHs: separating the neutrals from the ions

Observation	Proposal 7772, Observation 17: NGC2023-BKG-MIRI-MRS												Mon Aug 04 19:00:20 GMT 2025	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observation For: [NGC2023-MIRI-MRS (Obs 2)]													
Diagnostics	(Visit 17:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name			Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(9)	NGC2023-BKG			RA: 05 42 6.8400 (85.5285000d) Dec: -02 14 34.30 (-2.24286d) Equinox: J2000									
	Comments: Category=Calibration Description=[Telescope/sky background] Extended=NO													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction			
	F560W	All MRS			YES			FULL			Allow Auto Reorder			
Dithers	#	Dither Type			Optimized For			Direction						
	1	2-Point			EXTENDED SOURCE			NEGATIVE						
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F770W	FASTR1	5	1	1	Dither 1	2	2	27.75		
	1	SHORT(A)	MRSLONG		FASTR1	22	1	1	Dither 1	2	2	122.102		
	1	SHORT(A)	MRSSHORT		FASTR1	22	1	1	Dither 1	2	2	122.102		
	2		IMAGER	F1000W	FASTR1	5	1	1	Dither 1	2	2	27.75		
	2	MEDIUM(B)	MRSLONG		FASTR1	22	1	1	Dither 1	2	2	122.102		
	2	MEDIUM(B)	MRSSHORT		FASTR1	22	1	1	Dither 1	2	2	122.102		
	3		IMAGER	F1130W	FASTR1	5	1	1	Dither 1	2	2	27.75		
	3	LONG(C)	MRSLONG		FASTR1	22	1	1	Dither 1	2	2	122.102		
	3	LONG(C)	MRSSHORT		FASTR1	22	1	1	Dither 1	2	2	122.102		

Proposal 7772 - Observation 17 - Interstellar PAHs: separating the neutrals from the ions

Special Requirements	Sequence Observations 2, 17, Non-interruptible
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Proposal 7772 - Observation 4 - Interstellar PAHs: separating the neutrals from the ions

Observation	Proposal 7772, Observation 4: NGC1333-MIRI-MRS												Mon Aug 04 19:00:20 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[NGC1333-BKG-MIRI-MRS (Obs 18)]												
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(2)	NGC1333	RA: 03 29 10.4000 (52.2933333d) Dec: +31 21 51.12 (31.36420d) Equinox: J2000					Epoch of Position: 2015.5					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	Category=ISM												
	Description=[Reflection nebulae] Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	F560W	All MRS				YES			BRIGHTSKY		Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction			
	1	4-Point					EXTENDED SOURCE			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1		IMAGER	F1130W	FASTR1	5	2	1	Dither 1	4	8	38.072	
	1	LONG(C)	MRSLONG		FASTR1	12	1	1	Dither 1	4	4	133.202	213681
	1	LONG(C)	MRSSHORT		FASTR1	12	1	1	Dither 1	4	4	133.202	213681
	2		IMAGER	F1000W	FASTR1	5	2	1	Dither 1	4	8	38.072	
	2	MEDIUM(B)	MRSLONG		FASTR1	12	1	1	Dither 1	4	4	133.202	213681
	2	MEDIUM(B)	MRSSHORT		FASTR1	12	1	1	Dither 1	4	4	133.202	213681
	3		IMAGER	F770W	FASTR1	5	2	1	Dither 1	4	8	38.072	
	3	SHORT(A)	MRSLONG		FASTR1	12	1	1	Dither 1	4	4	133.202	213681
	3	SHORT(A)	MRSSHORT		FASTR1	12	1	1	Dither 1	4	4	133.202	213681

Proposal 7772 - Observation 4 - Interstellar PAHs: separating the neutrals from the ions

Special Requirements	Sequence Observations 4, 18, Non-interruptible
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Proposal 7772 - Observation 18 - Interstellar PAHs: separating the neutrals from the ions

Observation	Proposal 7772, Observation 18: NGC1333-BKG-MIRI-MRS												Mon Aug 04 19:00:20 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [NGC1333-MIRI-MRS (Obs 4)]												
Diagnosics	(Visit 18:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(10)	NGC1333-BKG	RA: 03 29 21.5100 (52.3396250d) Dec: +31 20 50.90 (31.34747d) Equinox: J2000										
	Comments: Category=Calibration Description=[Telescope/sky background] Extended=NO												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	F560W	All MRS				YES			BRIGHTSKY		Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction			
	1	2-Point					EXTENDED SOURCE			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1		IMAGER	F1130W	FASTR1	5	1	1	Dither 1	2	2	8.653	
	1	LONG(C)	MRSLONG		FASTR1	12	1	1	Dither 1	2	2	66.601	
	1	LONG(C)	MRSSHORT		FASTR1	12	1	1	Dither 1	2	2	66.601	
	2		IMAGER	F1000W	FASTR1	5	1	1	Dither 1	2	2	8.653	
	2	MEDIUM(B)	MRSLONG		FASTR1	12	1	1	Dither 1	2	2	66.601	
	2	MEDIUM(B)	MRSSHORT		FASTR1	12	1	1	Dither 1	2	2	66.601	
	3		IMAGER	F770W	FASTR1	5	1	1	Dither 1	2	2	8.653	
	3	SHORT(A)	MRSLONG		FASTR1	12	1	1	Dither 1	2	2	66.601	
	3	SHORT(A)	MRSSHORT		FASTR1	12	1	1	Dither 1	2	2	66.601	

Proposal 7772 - Observation 18 - Interstellar PAHs: separating the neutrals from the ions

Special Requirements	Sequence Observations 4, 18, Non-interruptible
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Proposal 7772 - Observation 5 - Interstellar PAHs: separating the neutrals from the ions

Observation	Proposal 7772, Observation 5: vdb133-NIRspec-IFU											Mon Aug 04 19:00:20 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Background Observations:[vdb133-BKG-NIRspec-IFU (Obs 6)]											
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(vdb133-NIRspec-IFU (Obs 5)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(3)	VDB133	RA: 20 31 4.2115 (307.7675479d) Dec: +36 57 22.67 (36.95630d) Equinox: J2000									
	Comments: Category=ISM Description=[Nebulae, Reflection nebulae] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Ex p	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395M/F290LP	NRSIRS2	8	1	false	true	NONE	4	4	2392.578	213682
	2	G395M/F290LP	NRSIRS2	8	1	true	true	NONE	4	4	2392.578	
Special Requirements	Sequence Observations 5, 6, Non-interruptible Same V3 PA 5, 7 (Aperture PAs differ)											

Proposal 7772 - Observation 6 - Interstellar PAHs: separating the neutrals from the ions

Observation	Proposal 7772, Observation 6: vdb133-BKG-NIRspec-IFU											Mon Aug 04 19:00:20 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Background Observation For: [vdb133-NIRspec-IFU (Obs 5)]											
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(4)	VDB133-BKG	RA: 20 31 16.7318 (307.8197158d) Dec: +36 50 42.01 (36.84500d) Equinox: J2000									
	Comments: Category=Calibration Description=[Telescope/sky background] Extended=NO											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395M/F290LP	NRSIRS2	8	1	false	true	NONE	4	4	2392.578	
Special Requirements	Sequence Observations 5, 6, Non-interruptible											

Proposal 7772 - Observation 7 - Interstellar PAHs: separating the neutrals from the ions

Observation	Proposal 7772, Observation 7: vdb133-MIRI-MRS												Mon Aug 04 19:00:20 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[vdb133-BKG-MIRI-MRS (Obs 8)]												
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
	(vdb133-MIRI-MRS (Obs 7)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(3)	VDB133	RA: 20 31 4.2115 (307.7675479d) Dec: +36 57 22.67 (36.95630d) Equinox: J2000										
	Comments: Category=ISM Description=[Nebulae, Reflection nebulae] Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction	
	F560W	All MRS				YES				FULL		Allow Auto Reorder	
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1		IMAGER	F770W	SLOWR1	6	6	1	Dither 1	4	24	3917.947	
	1	SHORT(A)	MRSLONG		SLOWR1	22	2	1	Dither 1	4	8	4300.186	254250
	1	SHORT(A)	MRSSHORT		SLOWR1	22	2	1	Dither 1	4	8	4300.186	254250
	2		IMAGER	F1000W	SLOWR1	6	6	1	Dither 1	4	24	3917.947	
	2	MEDIUM(B)	MRSLONG		SLOWR1	22	2	1	Dither 1	4	8	4300.186	254250
	2	MEDIUM(B)	MRSSHORT		SLOWR1	22	2	1	Dither 1	4	8	4300.186	254250
	3		IMAGER	F1130W	SLOWR1	6	6	1	Dither 1	4	24	3917.947	
	3	LONG(C)	MRSLONG		SLOWR1	22	2	1	Dither 1	4	8	4300.186	254250
	3	LONG(C)	MRSSHORT		SLOWR1	22	2	1	Dither 1	4	8	4300.186	254250

Proposal 7772 - Observation 7 - Interstellar PAHs: separating the neutrals from the ions

Special Requirements	Sequence Observations 7, 8, Non-interruptible Same V3 PA 5, 7 (Aperture PAs differ)
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Proposal 7772 - Observation 8 - Interstellar PAHs: separating the neutrals from the ions

Observation	Proposal 7772, Observation 8: vdb133-BKG-MIRI-MRS												Mon Aug 04 19:00:20 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [vdb133-MIRI-MRS (Obs 7)]												
Diagnosics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(4)	VDB133-BKG	RA: 20 31 16.7318 (307.8197158d) Dec: +36 50 42.01 (36.84500d) Equinox: J2000										
	Comments: Category=Calibration Description=[Telescope/sky background] Extended=NO												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	F560W	All MRS				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1		IMAGER	F770W	FASTR1	5	1	1	Dither 1	4	4	55.501	
	1	SHORT(A)	MRSLONG		SLOWR1	22	2	1	Dither 1	4	8	4300.186	
	1	SHORT(A)	MRSSHORT		SLOWR1	22	2	1	Dither 1	4	8	4300.186	
	2		IMAGER	F1000W	FASTR1	5	1	1	Dither 1	4	4	55.501	
	2	MEDIUM(B)	MRSLONG		SLOWR1	22	2	1	Dither 1	4	8	4300.186	
	2	MEDIUM(B)	MRSSHORT		SLOWR1	22	2	1	Dither 1	4	8	4300.186	
	3		IMAGER	F1130W	FASTR1	5	1	1	Dither 1	4	4	55.501	
	3	LONG(C)	MRSLONG		SLOWR1	22	2	1	Dither 1	4	8	4300.186	
	3	LONG(C)	MRSSHORT		SLOWR1	22	2	1	Dither 1	4	8	4300.186	

Proposal 7772 - Observation 8 - Interstellar PAHs: separating the neutrals from the ions

Special Requirements	Sequence Observations 7, 8, Non-interruptible
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Proposal 7772 - Observation 9 - Interstellar PAHs: separating the neutrals from the ions

Observation	Proposal 7772, Observation 9: vdb111-NIRspec-IFU											Mon Aug 04 19:00:20 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Background Observations:[vdb111-BKG-NIRspec-IFU (Obs 10)]											
Diagnostics	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(vdb111-NIRspec-IFU (Obs 9)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(5)	VDB111	RA: 17 18 52.3455 (259.7181063d)									
			Dec: +06 04 36.98 (6.07694d)									
			Equinox: J2000									
Template	Comments: Category=ISM Description=[Nebulae, Reflection nebulae] Extended=YES											
	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395M/F290LP	NRSIRS2	4	1	false	true	NONE	4	4	1225.467	213695
	2	G395M/F290LP	NRSIRS2	4	1	true	true	NONE	4	4	1225.467	
Special Requirements	Background Limited. Background no more than 10th percentile above minimum											
	Sequence Observations 9, 10, Non-interruptible Same V3 PA 9, 11 (Aperture PAs differ)											

Proposal 7772 - Observation 10 - Interstellar PAHs: separating the neutrals from the ions

Observation	Proposal 7772, Observation 10: vdb111-BKG-NIRspec-IFU											Mon Aug 04 19:00:20 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
	Background Observation For: [vdb111-NIRspec-IFU (Obs 9)]											
Diagnostics	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(6)	VDB111-BKG	RA: 17 19 11.4937 (259.7978904d) Dec: +06 05 47.71 (6.09659d) Equinox: J2000									
	Comments: Category=Calibration Description=[Telescope/sky background] Extended=NO											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395M/F290LP	NRSIRS2	4	1	false	true	NONE	4	4	1225.467	
Special Requirements	Sequence Observations 9, 10, Non-interruptible											

Proposal 7772 - Observation 11 - Interstellar PAHs: separating the neutrals from the ions

Observation	Proposal 7772, Observation 11: vdb111-MIRI-MRS												Mon Aug 04 19:00:20 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[vdb111-BKG-MIRI-MRS (Obs 12)]												
Diagnostics	(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
	(vdb111-MIRI-MRS (Obs 11)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(5)	VDB111	RA: 17 18 52.3455 (259.7181063d) Dec: +06 04 36.98 (6.07694d) Equinox: J2000										
	Comments: Category=ISM Description=[Nebulae, Reflection nebulae] Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray		Grating Wheel Direction	
	F560W	All MRS				YES				FULL		Allow Auto Reorder	
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1		IMAGER	F770W	SLOWR1	13	4	1	Dither 1	4	16	5255.782	
	1	SHORT(A)	MRSLONG		SLOWR1	30	2	1	Dither 1	4	8	5829.14	254593
	1	SHORT(A)	MRSSHORT		SLOWR1	30	2	1	Dither 1	4	8	5829.14	254593
	2		IMAGER	F1000W	SLOWR1	13	4	1	Dither 1	4	16	5255.782	
	2	MEDIUM(B)	MRSLONG		SLOWR1	30	2	1	Dither 1	4	8	5829.14	254593
	2	MEDIUM(B)	MRSSHORT		SLOWR1	30	2	1	Dither 1	4	8	5829.14	254593
	3		IMAGER	F1130W	SLOWR1	13	4	1	Dither 1	4	16	5255.782	
	3	LONG(C)	MRSLONG		SLOWR1	30	2	1	Dither 1	4	8	5829.14	254593
	3	LONG(C)	MRSSHORT		SLOWR1	30	2	1	Dither 1	4	8	5829.14	254593

Proposal 7772 - Observation 11 - Interstellar PAHs: separating the neutrals from the ions

Special Requirements	Background Limited. Background no more than 10th percentile above minimum Sequence Observations 11, 12, Non-interruptible Same V3 PA 9, 11 (Aperture PAs differ)
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Proposal 7772 - Observation 12 - Interstellar PAHs: separating the neutrals from the ions

Observation	Proposal 7772, Observation 12: vdb111-BKG-MIRI-MRS												Mon Aug 04 19:00:20 GMT 2025	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observation For: [vdb111-MIRI-MRS (Obs 11)]													
Diagnostics	(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name			Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(6)	VDB111-BKG			RA: 17 19 11.4937 (259.7978904d) Dec: +06 05 47.71 (6.09659d) Equinox: J2000									
	Comments: Category=Calibration Description=[Telescope/sky background] Extended=NO													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction			
	F560W	All MRS			YES			FULL			Allow Auto Reorder			
Dithers	#	Dither Type					Optimized For			Direction				
	1	4-Point					EXTENDED SOURCE			NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F770W	FASTR1	5	1	1	Dither 1	4	4	55.501		
	1	SHORT(A)	MRSLONG		SLOWR1	30	2	1	Dither 1	4	8	5829.14		
	1	SHORT(A)	MRSSHORT		SLOWR1	30	2	1	Dither 1	4	8	5829.14		
	2		IMAGER	F1000W	FASTR1	5	1	1	Dither 1	4	4	55.501		
	2	MEDIUM(B)	MRSLONG		SLOWR1	30	2	1	Dither 1	4	8	5829.14		
	2	MEDIUM(B)	MRSSHORT		SLOWR1	30	2	1	Dither 1	4	8	5829.14		
	3		IMAGER	F1130W	FASTR1	5	1	1	Dither 1	4	4	55.501		
	3	LONG(C)	MRSLONG		SLOWR1	30	2	1	Dither 1	4	8	5829.14		
	3	LONG(C)	MRSSHORT		SLOWR1	30	2	1	Dither 1	4	8	5829.14		

Proposal 7772 - Observation 12 - Interstellar PAHs: separating the neutrals from the ions

Special Requirements	Sequence Observations 11, 12, Non-interruptible
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Proposal 7772 - Observation 13 - Interstellar PAHs: separating the neutrals from the ions

Observation	Proposal 7772, Observation 13: vdb101-NIRspec-IFU Mon Aug 04 19:00:20 GMT 2025											
	Diagnostic Status: Warning Observing Template: NIRSpec IFU Spectroscopy Background Observations:[vdb101-BKG-NIRspec-IFU (Obs 14)]											
Diagnostics	(Visit 13:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(7)	VDB101	RA: 16 19 2.2506 (244.7593775d) Dec: -20 14 32.10 (-20.24225d) Equinox: J2000									
Template	Comments: Category= <i>ISM</i> Description=[<i>Nebulae, Reflection nebulae</i>] Extended= <i>YES</i>											
	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395M/F290LP	NRSIRS2	5	1	false	true	NONE	4	4	1517.245	214759
	2	G395M/F290LP	NRSIRS2	5	1	true	true	NONE	4	4	1517.245	
Special Requirements	Background Limited. Background no more than 10th percentile above minimum											
	Sequence Observations 13, 14, Non-interruptible											

Proposal 7772 - Observation 14 - Interstellar PAHs: separating the neutrals from the ions

Observation	Proposal 7772, Observation 14: vdb101-BKG-NIRspec-IFU Mon Aug 04 19:00:20 GMT 2025											
	Diagnostic Status: Warning Observing Template: NIRSpec IFU Spectroscopy Background Observation For: [vdb101-NIRspec-IFU (Obs 13)]											
Diagnostics	(Visit 14:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections		Miscellaneous			
	(8)	VDB101-BKG	RA: 16 19 36.0675 (244.9002813d) Dec: -20 18 47.83 (-20.31329d) Equinox: J2000									
Template	<i>Comments:</i> <i>Category=Calibration</i> <i>Description=[Telescope/sky background]</i> <i>Extended=NO</i>											
	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395M/F290LP	NRSIRS2	5	1	false	true	NONE	4	4	1517.245	
Special Requirements	Sequence Observations 13, 14, Non-interruptible											

Proposal 7772 - Observation 15 - Interstellar PAHs: separating the neutrals from the ions

Observation	Proposal 7772, Observation 15: vdb101-MIRI-MRS												Mon Aug 04 19:00:20 GMT 2025	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[vdb101-BKG2-MIRI-MRS (Obs 16)]													
Diagnostics	(Visit 15:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name			Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(11)	VDB101-visit-15-copy			RA: 16 19 2.2506 (244.7593775d) Dec: -20 14 32.10 (-20.24225d) Equinox: J2000									
	Comments: Category=ISM Description=[Nebulae, Reflection nebulae] Extended=YES													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction			
	F560W	All MRS			YES			FULL			Allow Auto Reorder			
Dithers	#	Dither Type					Optimized For			Direction				
	1	4-Point					EXTENDED SOURCE			NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F770W	SLOWR1	13	4	1	Dither 1	4	16	5255.782		
	1	SHORT(A)	MRSLONG		SLOWR1	30	2	1	Dither 1	4	8	5829.14	254595	
	1	SHORT(A)	MRSSHORT		SLOWR1	30	2	1	Dither 1	4	8	5829.14	254595	
	2		IMAGER	F1000W	SLOWR1	13	4	1	Dither 1	4	16	5255.782		
	2	MEDIUM(B)	MRSLONG		SLOWR1	30	2	1	Dither 1	4	8	5829.14	254595	
	2	MEDIUM(B)	MRSSHORT		SLOWR1	30	2	1	Dither 1	4	8	5829.14	254595	
	3		IMAGER	F1130W	SLOWR1	13	4	1	Dither 1	4	16	5255.782		
	3	LONG(C)	MRSLONG		SLOWR1	30	2	1	Dither 1	4	8	5829.14	254595	
	3	LONG(C)	MRSSHORT		SLOWR1	30	2	1	Dither 1	4	8	5829.14	254595	

Proposal 7772 - Observation 15 - Interstellar PAHs: separating the neutrals from the ions

Special Requirements	Background Limited. Background no more than 10th percentile above minimum Sequence Observations 15, 16, Non-interruptible
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Proposal 7772 - Observation 16 - Interstellar PAHs: separating the neutrals from the ions

Observation	Proposal 7772, Observation 16: vdb101-BKG2-MIRI-MRS												Mon Aug 04 19:00:20 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [vdb101-MIRI-MRS (Obs 15)]												
Diagnostics	(Visit 16:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(12)	VDB101-BKG-visit-15-copy	RA: 16 20 51.8165 (245.2159021d) Dec: -20 27 42.07 (-20.46169d) Equinox: J2000										
	Comments: Category=Calibration Description=[Telescope/sky background] Extended=NO												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	F560W	All MRS				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For			Direction			
	1	4-Point					EXTENDED SOURCE			NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1		IMAGER	F770W	FASTR1	5	1	1	Dither 1	4	4	55.501	
	1	SHORT(A)	MRSLONG		SLOWR1	30	2	1	Dither 1	4	8	5829.14	
	1	SHORT(A)	MRSSHORT		SLOWR1	30	2	1	Dither 1	4	8	5829.14	
	2		IMAGER	F1000W	FASTR1	5	1	1	Dither 1	4	4	55.501	
	2	MEDIUM(B)	MRSLONG		SLOWR1	30	2	1	Dither 1	4	8	5829.14	
	2	MEDIUM(B)	MRSSHORT		SLOWR1	30	2	1	Dither 1	4	8	5829.14	
	3		IMAGER	F1130W	FASTR1	5	1	1	Dither 1	4	4	55.501	
	3	LONG(C)	MRSLONG		SLOWR1	30	2	1	Dither 1	4	8	5829.14	
	3	LONG(C)	MRSSHORT		SLOWR1	30	2	1	Dither 1	4	8	5829.14	

Proposal 7772 - Observation 16 - Interstellar PAHs: separating the neutrals from the ions

Special Requirements	Sequence Observations 15, 16, Non-interruptible
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